

Subject Description Form

Subject Code	ITC3226K
Subject Title	Knitwear Technology I
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Co-requisite: ITC3222K Knitted Structure Design and ITC3227K Knitwear Technology II Exclusion: SFT317KD Knitwear Technology
Objectives	The subject provides an outline of knitwear manufacturing based on the electronic power knitting technique. It includes the use of materials, basic knitted structures and garment styles, and the methods of garment assemble accordingly.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) classify and analyse the materials used in knitwear production; (b) define and explain the essential terms in fabrics/panels manufactured by electronic flat knitting techniques; (c) demonstrate and integrate design and technical skills of computer design software and electronic flat knitting machine for creating different stitches and knitted structures; (d) communicate effectively with industrial practitioners; (e) identify and apply the safety and ethical practices in the industry.
Subject Synopsis/ Indicative Syllabus	(I) Electronic Flat Knitting Machines and Suitable Yarn for Knitwear Specification, configuration, operation and safety instructions for electronic flat knitting machines. Basic knitted structures, including single jersey, ribs, double jersey, purl, lace, cable and standard double Jacquard. (II) Computer Design Software for Knitwear Drawing tools and colour-code for elementary knitting structures. Option lines for knitting process control, including yarn carrier, stitch length and fabric take-down tension. Compiling of knitting program and fabric simulation.

Subject Synopsis/ Indicative Syllabus	(III) Knitting Techniques Advanced stitch development method (package method) Basics of fully fashion knitting. (IV) Developing Skills for Knitted Structures Mix and combination of yarn, basic fabric structure and knitting technique. Creative fabric structure design project. (V) Safety and Ethical Practices in the Industry General safety and general ethical practices when operating the machines in knitwear industry							
Teaching/Learning Methodology	Lectures will be conducted to provide background knowledge of specific computer design software and electronic flat knitting machines for knitwear production. More time will be spent in the knitting laboratory for demonstration and hands-on experience aiming to provide a solid understanding of knitted structures and designs. An elementary stitch design project will form the core assignment serving as a student-centered learning platform.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	Continuous Assessment	100%	✓	✓	✓	✓	✓	
	1. Assignments	60%	✓	✓				
	2. Project	40%	✓	✓	✓	✓	✓	
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments will assess students’ understanding on principles of electronic flat knitting techniques and fabric structures. Individual project will assess students’ manipulative skills and creativities in more comprehensive means.							

Student Study Effort Expected	Class contact:	
	• Workshop	39 Hrs.
	Other student study effort:	
	• Assignment	36 Hrs.
	• Project	30 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	<u>Recommended Textbooks</u> Brackenbury, T. (1992), <i>Knitted Clothing Technology</i>. Blackwell Scientific Publications, UK. Gligorijevic, V. (2016), <i>Technology of knitting: theoretical and experimental analysis</i>. LAP Lambert Academic Publishing. Raz, S. (1993), <i>Flat knitting technology</i>. Universal Maschienfabrik, Westhausen. Spencer, D. (2001), <i>Knitting technology a comprehensive handbook and practical guide</i>. 3rd Edition, Woodhead publishing Ltd, Cambridge, UK. Tellier-Loumagne, F. (2005), <i>The art of knitting: Inspirational stitches, textures and surfaces</i>. Thames & Hudson, London. <u>Electronic Resources</u> WGSN www.wgsn.com Qiliabc www.qiliabc.com www.shimaseiki.com www.stoll.de Help file for computer design software (shimaseiki, stoll)	